

Roll No:- 2023101108

Varun Gupta

LPDE AssignmentA1Given: Conservative^{central} Force

$$\mathbf{F} = F(r) \hat{r}$$

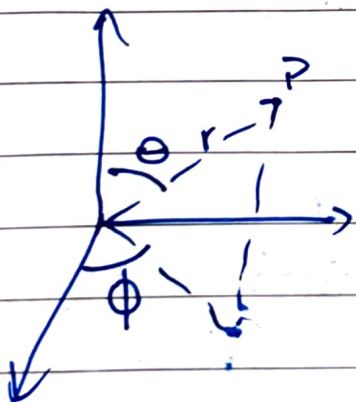
$$\text{Now, } \mathbf{F} = -\frac{\partial V}{\partial \mathbf{r}}$$

Let a particle P in 3D space. The sph. coordinates (r, θ, ϕ) are defined wrt Cartesian frame (x, y, z) as :

$r \rightarrow$ radial distance from origin

$\theta \rightarrow$ polar (colatitude) ang from $+z$.

$\phi \rightarrow$ azimuthal angle from $+x$ in xy



$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

Also inversely,

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}(z/r)$$

$$\phi = \tan^{-1}(y/x)$$

$$\mathbf{r} = r \hat{\mathbf{r}}$$

Now,

$$\mathbf{r} = x \hat{\mathbf{i}} + y \hat{\mathbf{j}} + z \hat{\mathbf{k}}$$

Subst our transf,

$$\mathbf{r} = (r \sin \theta \cos \phi) \hat{\mathbf{i}} + (r \sin \theta \sin \phi) \hat{\mathbf{j}} + (r \cos \theta) \hat{\mathbf{k}}$$

To find unit Basis vectors,

$$\hat{\mathbf{e}}_i = \frac{\partial \mathbf{r}}{\partial q_i} / \left| \frac{\partial \mathbf{r}}{\partial q_i} \right|$$

Now,

Radial: $\frac{\partial \mathbf{r}}{\partial r} = \sin \theta \cos \phi \hat{\mathbf{i}} + \sin \theta \sin \phi \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}}$

with magnitude = 1

$$\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{i}} + \sin \theta \sin \phi \hat{\mathbf{j}} + \cos \theta \hat{\mathbf{k}}$$

Polar: Magnitude = r

$$\begin{aligned} \hat{\theta} &= \frac{1}{r} \left(r \cos \theta \cos \phi \hat{\mathbf{i}} + r \cos \theta \sin \phi \hat{\mathbf{j}} - r \sin \theta \hat{\mathbf{k}} \right) \\ &= \cos \theta \cos \phi \hat{\mathbf{i}} + \cos \theta \sin \phi \hat{\mathbf{j}} - \sin \theta \hat{\mathbf{k}} \end{aligned}$$

Azimuthal,

$$\frac{\partial \mathbf{r}}{\partial \phi} = -r \sin \theta \sin \phi \hat{i} + r \sin \theta \cos \phi \hat{j}$$

$$\text{Magnitude} = r \sin \theta$$

$$\hat{\phi} = -\sin \phi \hat{i} + \cos \phi \hat{j}$$

$\Rightarrow$ we can check these vectors orthonormal

Now, to find velocity,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}$$

$$\mathbf{r} = r \hat{r}$$

$$\mathbf{v} = \dot{r} \hat{r} + r \dot{\hat{r}}$$

Since sph. basis vector changes dirⁿ as Particle moves, their time derivative non zero

$$\therefore, \frac{d\hat{r}}{dt} = \frac{\partial \hat{r}}{\partial \theta} \dot{\theta} + \frac{\partial \hat{r}}{\partial \phi} \dot{\phi}$$

$$\frac{\partial \hat{r}}{\partial \theta} = \cos \theta \cos \phi \hat{i} + \cos \theta \sin \phi \hat{j} = -\sin \theta \hat{r}$$

$$\frac{\partial \hat{r}}{\partial \phi} = -\sin \theta \sin \phi \hat{i} + \sin \theta \cos \phi \hat{j} = \sin \theta \hat{\phi}$$

$$\dot{\hat{r}} = \dot{\theta} \hat{\theta} + \dot{\phi} (\sin \theta) \hat{\phi}$$

$$\begin{aligned}
 \mathbf{v} &= \dot{r} \hat{r} + r \dot{\theta} \hat{\theta} + r \sin \theta \dot{\phi} \hat{\phi} \\
 &= (\dot{r}) \hat{r} + (r \dot{\theta}) \hat{\theta} + (r \dot{\phi} \sin \theta) \hat{\phi}
 \end{aligned}$$

$$\begin{array}{ccccc}
 \mathbf{v} = v_r \hat{r} + v_\theta \hat{\theta} + v_\phi \hat{\phi} & & & & \\
 \downarrow & & \downarrow & & \downarrow \\
 \dot{r} & & r \dot{\theta} & & r \dot{\phi} \sin \theta
 \end{array}$$

A2 $L = T - V$

Now, $T = \frac{1}{2} m \mathbf{v}^2$

$$\mathbf{v}^2 = \mathbf{v} \cdot \mathbf{v}$$

$$= \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

(BC2 basis vector are orthonormal)

$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2)$$

Let assume a general potential field
 $V(r, \theta, \phi) = V(r)$ [Since Force is Central]

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) - V(r)$$

By Principle of Least action,

$$S = \int_{t_1}^{t_2} L(q_i, \dot{q}_i, t) dt$$

Using $\delta S = 0$

By Euler Lagrange Equation,

For each gen. coordinate $q_i \in \{r, \theta, \phi\}$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$$

(i) For radial (r)

$$\frac{\partial L}{\partial \dot{r}} = m \dot{r} \longrightarrow \frac{d}{dt} (m \dot{r}) = m \ddot{r}$$

$$\frac{\partial L}{\partial r} = m r \dot{\theta}^2 + m r \sin^2 \theta \dot{\phi}^2 - \frac{\partial V}{\partial r}$$

$$m \ddot{r} - (m r \dot{\theta}^2 + m r \sin^2 \theta \dot{\phi}^2 - \frac{\partial V}{\partial r}) = 0$$

$$\boxed{m \ddot{r} - m r \dot{\theta}^2 - m r \sin^2 \theta \dot{\phi}^2 = - \frac{\partial V}{\partial r}}$$

" $\downarrow$
 $f(r)$

(ii) For Polar Angle

$$\frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}$$

$$\frac{d}{dt}(m r^2 \dot{\theta}) = m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta}$$

$$\frac{\partial L}{\partial \theta} = m r^2 \sin \theta \cos \theta \dot{\phi}^2 - \frac{\partial V}{\partial \theta} \rightarrow 0$$

$$m r^2 \ddot{\theta} + 2 m r \dot{r} \dot{\theta} - m r^2 \sin \theta \cos \theta \dot{\phi}^2 = 0$$

(iii) For Azimuthal Angle

$$\frac{\partial L}{\partial \dot{\phi}} = m r^2 \sin^2 \theta \dot{\phi} \quad (L_z)$$

$$\frac{d}{dt}(m r^2 \sin^2 \theta \dot{\phi}) = \frac{\partial L}{\partial \phi} = \frac{\partial V}{\partial \phi} = 0$$

$$m r^2 \sin^2 \theta \dot{\phi} = k$$

A3

Hamiltonian Formalism $\rightarrow$

$$p_{q_i} = \frac{\partial L}{\partial \dot{q}_i}$$

$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \sin^2 \theta \dot{\phi}^2 + r^2 \dot{\theta}^2) - V(r)$$

$$P_r = \frac{\partial L}{\partial \dot{r}} = m\dot{r} \quad \text{--- (1) } (\because V = \text{fn of } r)$$

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \quad \text{--- (2)}$$

$$P_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2 \sin^2\theta \dot{\phi} \quad \text{--- (3)}$$

$$H_{\text{system}} = \sum P_i \dot{q}_i - L$$

$$= P_r \dot{r} + P_\theta \dot{\theta} + P_\phi \dot{\phi} - L$$

$$= m\dot{r}^2 + mr^2\dot{\theta}^2 + mr^2 \sin^2\theta \dot{\phi}^2 - \frac{1}{2}m[\dot{r}^2 + r^2 \sin^2\theta \dot{\phi}^2 + r^2 \dot{\theta}^2] + V(r)$$

$$\left\{ H_{\text{sys}} = \frac{1}{2}m(\dot{r}^2 + r^2 \sin^2\theta \dot{\phi}^2 + r^2 \dot{\theta}^2) + V(r) \right.$$

$$H_{\text{sys}} = \frac{P_r^2}{2m} + \frac{P_\theta^2}{2mr^2} + \frac{P_\phi^2}{2mr^2 \sin^2\theta} + V(r)$$

(Using 1, 2, 3)

By Hamiltonian's canonical eqn $\rightarrow$

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = - \frac{\partial H}{\partial q_i}$$

$$\dot{r} = \frac{\partial H}{\partial p_r}$$

$\Rightarrow$ For Radial Coordinate (r, p_r)

$$\dot{r} = \frac{\partial H}{\partial p_r} = \frac{p_r}{m}$$

$$\dot{p}_r = - \frac{\partial H}{\partial r} = \frac{p_\theta^2}{mr^3} + \frac{p_\phi^2}{mr^3 \sin^2 \theta} - \frac{\partial V(r)}{\partial r}$$

$\Rightarrow$ For Polar (θ, p_θ)

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{mr^2}$$

$$\dot{p}_\theta = - \frac{\partial H}{\partial \theta} = \frac{p_\phi^2 \cos \theta}{mr^2 \sin^3 \theta}$$

$\Rightarrow$ For Azimuthal (ϕ, p_ϕ)

$$\dot{\phi} = \frac{\partial H}{\partial p_\phi} = \frac{p_\phi}{mr^2 \sin^2 \theta}$$

$$\dot{p}_\phi = - \frac{\partial H}{\partial \phi} = 0 \quad (p_\phi = \text{const})$$

Ans In central Force field, the pot. energy $V(r)$ depends only on the distance from the origin. This arrangement provide sym. according to Noether's Th. lead to conservation of following constant of Motions:

(i) P_ϕ

$$H = \frac{P_r^2}{2m} + \frac{P_\theta^2}{2mr^2} + \frac{P_\phi^2}{2mr^2 \sin^2 \theta} + V(r)$$

$$\frac{\partial H}{\partial \phi} = 0$$

$$\dot{P}_\phi = 0 \Rightarrow P_\phi = \text{Constant}$$

This is Rotational Invariance about z axis.
 L_z is conserved $\Rightarrow$ no torque in z axis

(ii) Total Angular Momentum (L^2)

$$L = r \times p$$

(For central Force

$$L^2 = P_\theta^2 + \frac{P_\phi^2}{\sin^2 \theta}$$

the L is conserved)

If we calculate Poisson bracket $\{L^2, H\} = 0 \Rightarrow L^2 = \text{Constant of Motion}$

It arise from full 3D Rot Sym. It implies position vector $\mathbf{r} \perp \mathbf{L}$ always stay in a fixed plane $\perp$ to $\mathbf{L}$.

By which we can simplify 3D problem in 2D polar with $\theta = \pi/2$

(iii) Total Energy.

H we derived is Total Energy (T+V). Since Trans. are independent of time & Pot is not explicitly time dep.

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} = 0 \Rightarrow E = \text{constant}$$

This is conseq. of time translational Symmetry.

(iv) The conservation of L allow us to rewrite radial eq. of Motion just using r . By subst. $p_\phi \rightarrow L$ in Ham., we define $V_{\text{eff}}(r)$

$$H^2 = \frac{p_r^2}{2m} + \left(\frac{L^2}{2mr^2} + V(r) \right) = \frac{p_r^2}{2m} + V_{\text{eff}}(r)$$

Now, $\frac{L^2}{2mr^2} \rightarrow$ Repulsive pot. Arise Purely

from geometry of rot. & prevents particle from reaching origin.
(unless $L=0$)

∴, The motion of Particle can be stated as 1D Prob. problem.